IGCSE 0607 Practice Questions — E3.1

Graphs of functions

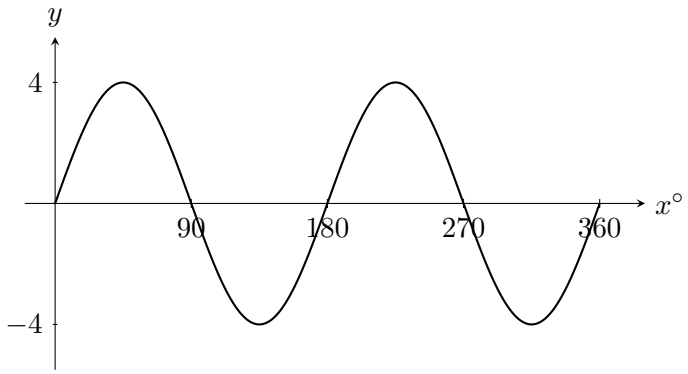
Total Questions: 42 — **Total Marks:** 233 Marks

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- **E3.1** Graphs of functions
  → **Paper 2 (Non-Calculator)** *(22 Qs)* · **Paper 4 (Calculator)** *(20 Qs)*

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1 The diagram shows the graph of $y = a\sin(bx)$ for $0° \leq x \leq 360°$.



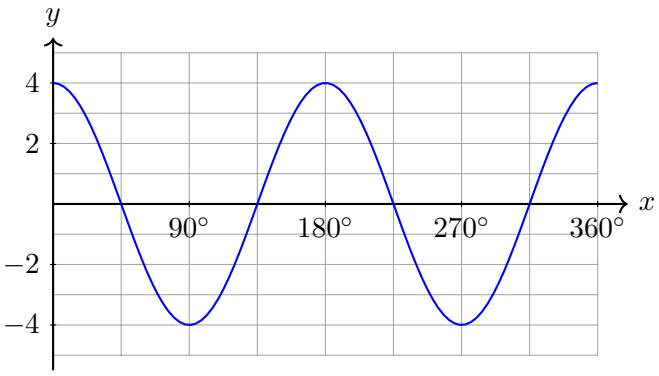
(a) Find the value of a.

$a =$ ..... **[1]**

(b) Find the value of b.

$b =$ ..... **[1]**

2 The diagram shows a sketch of the graph of $y = a\cos(bx)$ for $0° \leq x \leq 360°$.



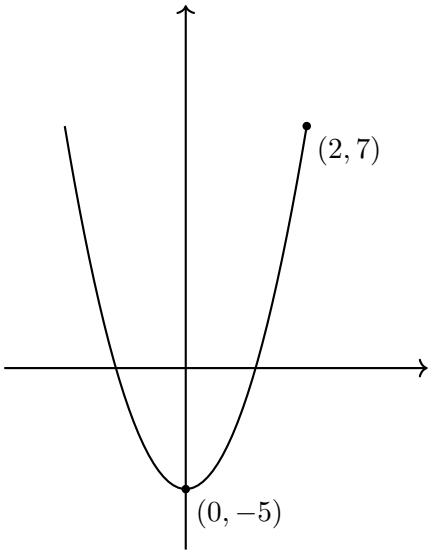
(a) Write down the value of a.

$a =$ ..... **[1]**

(b) Find the value of b.

$b =$ ..... **[2]**

3 The diagram shows a sketch of the graph of the function $f(x) = ax^2 + c$.
The graph passes through the points $(0, -5)$ and $(2, 7)$.



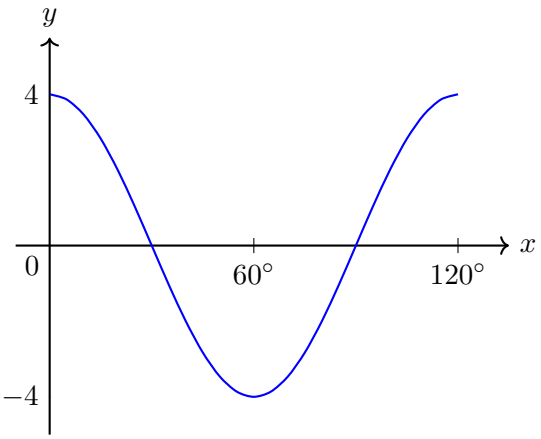
(a) Write down the value of c.

$c =$ ..... **[1]**

(b) Find the value of a.

$a =$ ..... **[2]**

4 The diagram shows the sketch of the graph of $y = a\cos(bx)$ for $0° \leq x \leq 120°$.



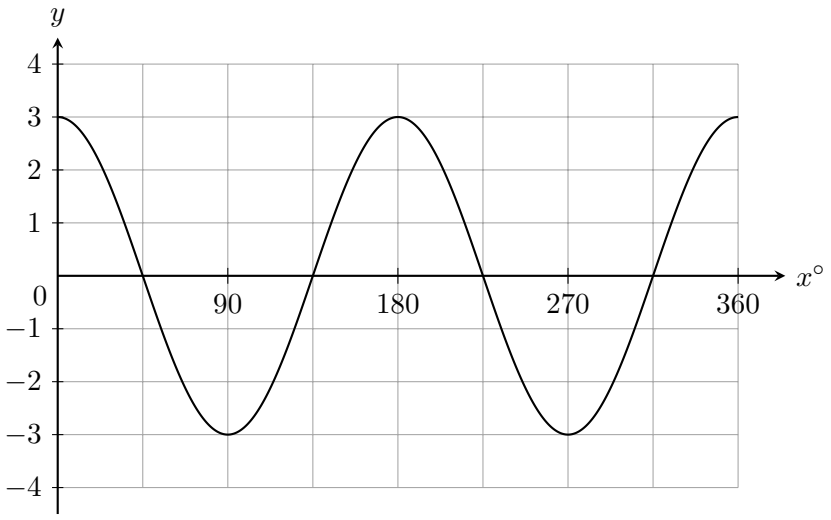
(a) Write down the amplitude of the graph.

..... **[1]**

(b) Find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[2]**

5 The diagram shows the graph of the trigonometric function $f(x) = a\cos(bx)$ for $0° \leq x \leq 360°$.



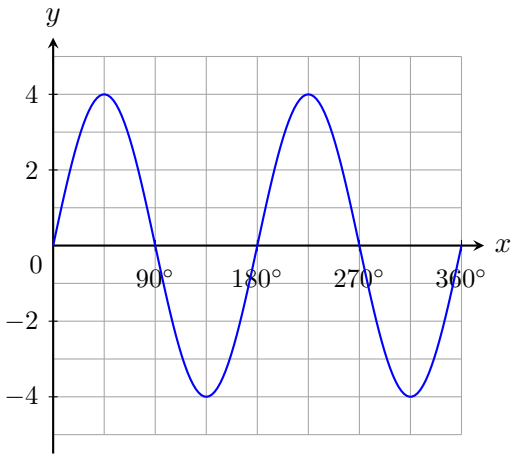
(a) Write down the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[2]**

(b) Write down the coordinates of the first minimum point where $x > 0$.

(..... , ...) **[1]**

6 The diagram shows the graph of $y = a\sin(bx)$ for $0° \leq x \leq 360°$.



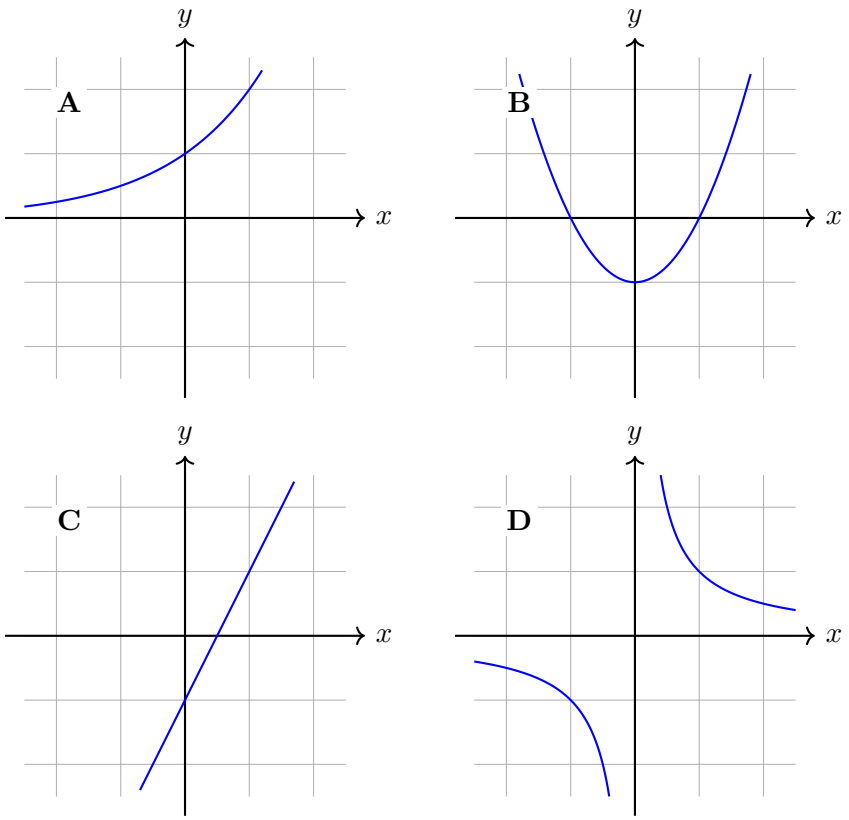
(a) Find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[2]**

(b) Write down the period of the function.

..... ° **[1]**

7 Four graphs are shown below, labelled A, B, C, and D.



(a) Write down the letter of the graph that represents each function.

(i)

$$y = x^2 - 1$$

..... **[1]**

(ii)

$$y = 2^x$$

..... **[1]**

(iii)

$$y = \frac{1}{x}$$

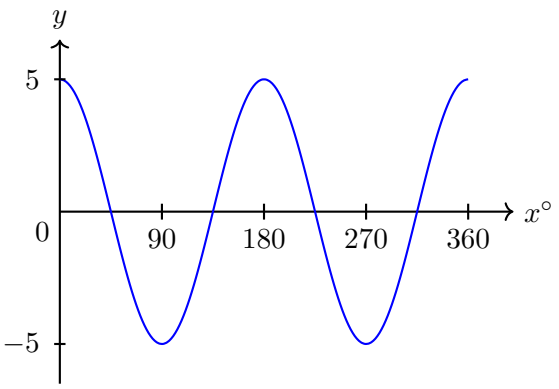
..... **[1]**

(iv)

$$y = 2x - 1$$

..... **[1]**

8 The diagram shows the graph of $y = a\cos(bx)$ for $0° \leq x \leq 360°$.



(a) Write down the amplitude of the graph.

..... **[1]**

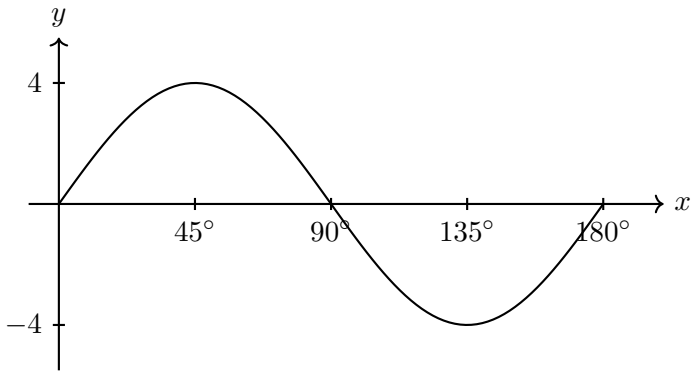
(b) Write down the period of the graph.

..... ° **[1]**

(c) Find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[2]**

9 The diagram shows the graph of $y = a\sin(bx)$ for $0° \leq x \leq 180°$.



(a) State the amplitude.

..... **[1]**

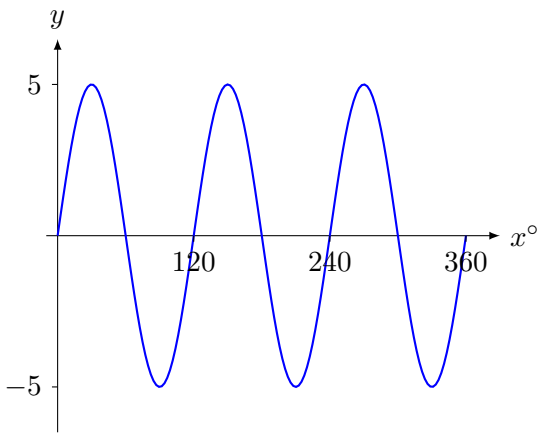
(b) State the period of the function.

..... ° **[1]**

(c) Write down the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[2]**

10 The diagram shows the graph of $y = a\sin(bx)$ for $0° \leq x \leq 360°$.



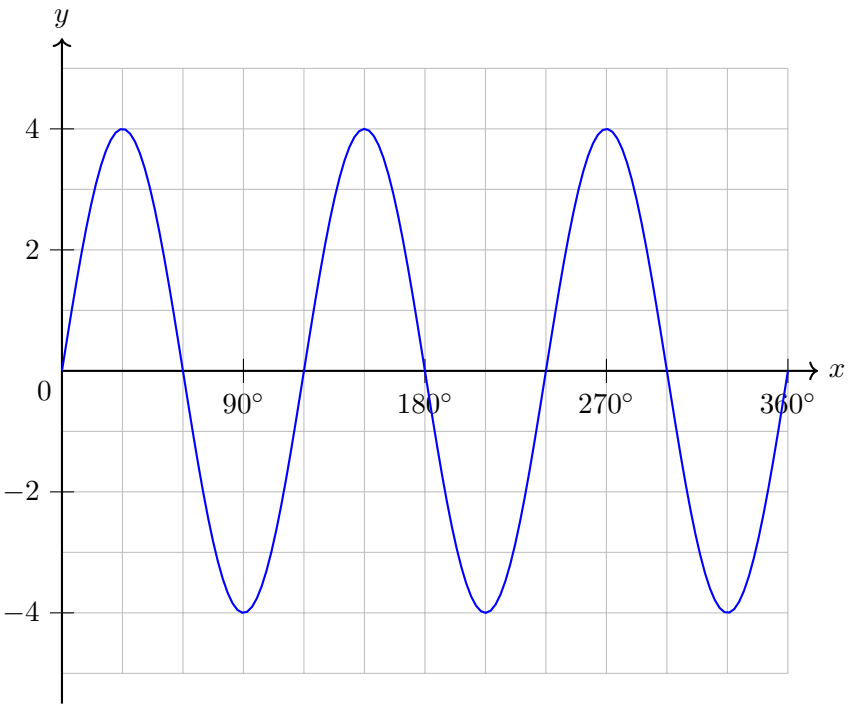
(a) Find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[2]**

(b) The graph of $y = kx^3$ passes through the point $(2, -24)$.
Find the value of k.

$k =$ ..... **[2]**

11 The diagram shows the graph of $y = a\sin(bx)$ for $0° \leq x \leq 360°$.
The values of a and b are positive integers.



(a) Find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[2]**

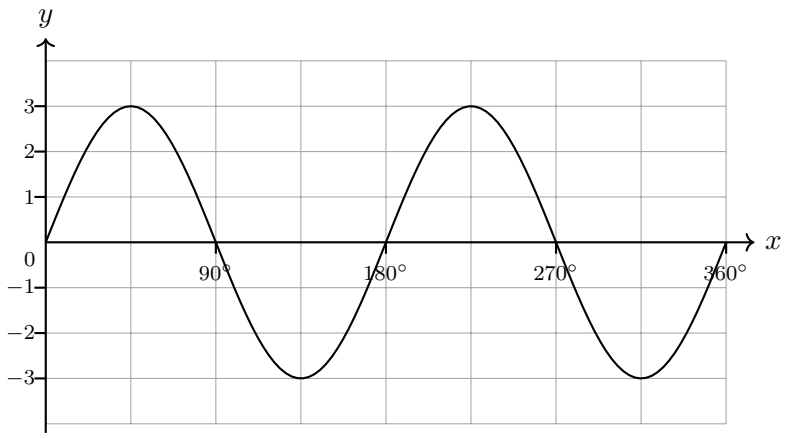
(b) Write down the period of the function.

..... ° **[1]**

(c) State the number of solutions to the equation $a\sin(bx) = 2$ in the interval $0° \leq x \leq 360°$.

..... **[1]**

12 The diagram shows the graph of the trigonometric function $y = a\sin(bx)$ for $0° \leq x \leq 360°$, where a and b are positive constants.



(a) Write down the amplitude and the period of the function.

Amplitude = ... , Period = ... **[2]**

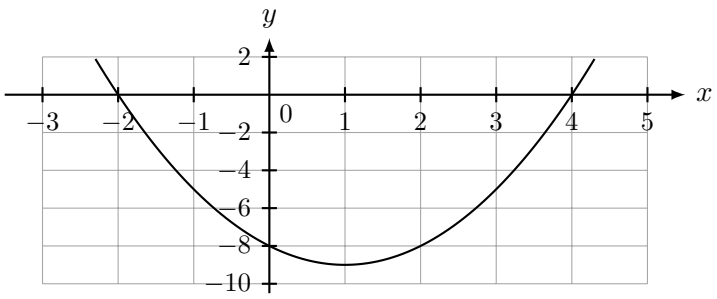
(b) Determine the values of a and b.

a = ... , b = ... **[2]**

(c) Use the graph to state the number of solutions to the equation $a\sin(bx) = -2$ for $0° \leq x \leq 360°$.

..... **[1]**

13 The diagram shows the graph of the function $y = x^2 + bx + c$.



(a) Write down the value of c.

$c =$ ..... **[1]**

(b) Find the value of b.

$b =$ ..... **[2]**

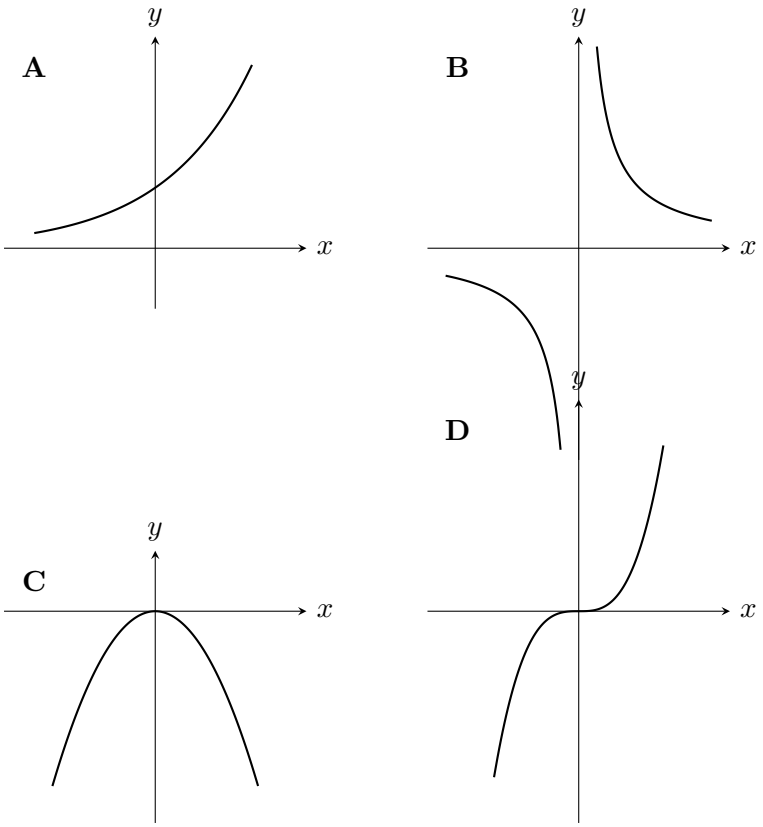
(c) Write down the coordinates of the minimum point of the graph.

(..... , ...) **[2]**

14 Six equations are shown below.

$$y = 2^x \qquad y = x^2 \qquad y = -x^2 \qquad y = x^3 \qquad y = \frac{1}{x} \qquad y = \cos x$$

The diagrams show four of these graphs, labelled A, B, C and D.

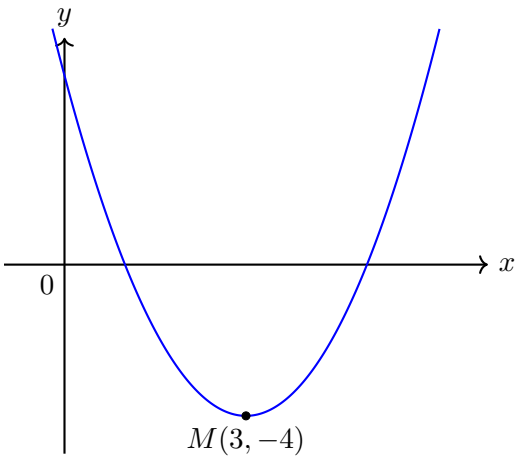


Write down the correct equation for each graph.

$A =$ ..... , $B =$ ..... , $C =$ ..... , $D =$ .....

[4]

15 The diagram shows a sketch of the graph of $y = x^2 + bx + c$.
The minimum point on the graph is $M(3, -4)$.



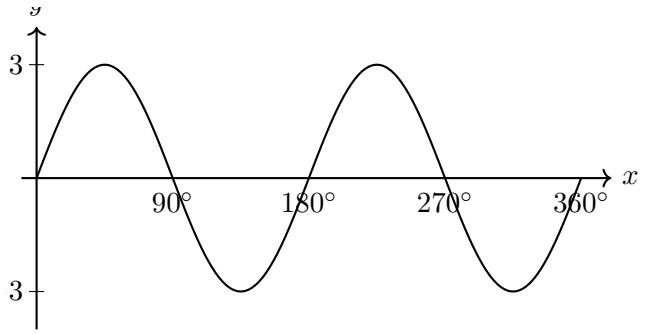
(a) Work out the value of b and the value of c.

$b =$ ..... , $c =$ ..... **[3]**

(b) Find the coordinates of the points where the graph crosses the x-axis.

..... **[2]**

16 The diagram shows a sketch of the graph of $f(x) = a\sin(bx)$ for $0° \leq x \leq 360°$.



(a) Write down the value of a.

$a =$ ..... **[1]**

(b) Write down the value of b.

$b =$ ..... **[1]**

(c) Solve the equation $f(x) = 1.5$ for $0° \leq x \leq 180°$.

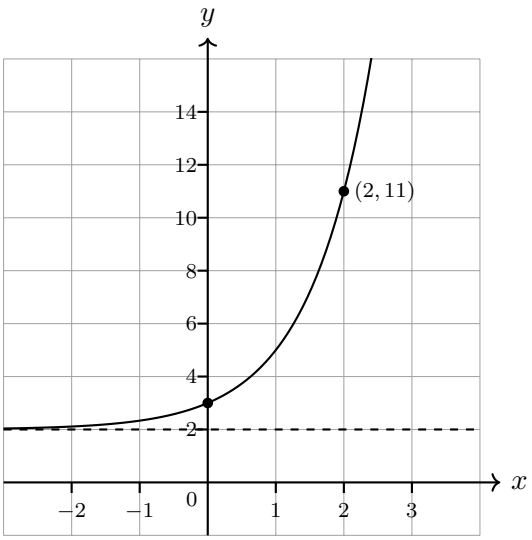
$x =$ ..... **[3]**

17 The graph of the quadratic function $f(x) = ax^2 + bx + c$ has a line of symmetry at $x = 2$.
The graph passes through the points $(0, -5)$ and $(2, 3)$.

Find the values of a, b and c.

$a =$ ..... , $b =$ ..... , $c =$ ..... **[5]**

18 The diagram shows the graph of $f(x) = a^x + c$, where $a > 0$ and c are constants.
The curve passes through the points $(0, 3)$ and $(2, 11)$, and has a horizontal asymptote at $y = 2$.



(a) Find the values of a and c.

$a =$ ..... , $c =$ ..... **[3]**

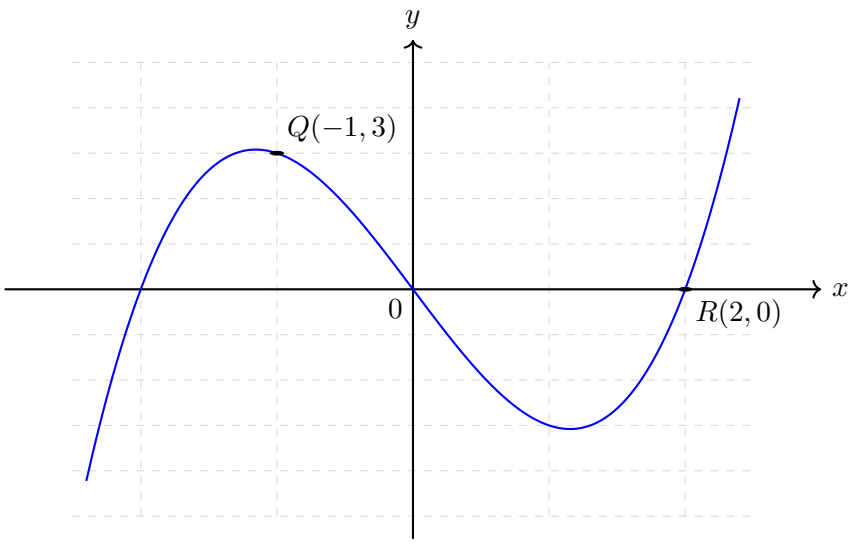
(b) Find the value of $f(-1)$, giving your answer as a fraction.

$f(-1) =$ ..... **[1]**

(c) State the range of the function $f(x)$.

..... **[2]**

19 The diagram shows a sketch of the curve with equation $y = x^3 + ax^2 + bx$.
The curve passes through the points $Q(-1, 3)$ and $R(2, 0)$.



(a) Use the coordinates of Q and R to form two simultaneous equations in terms of a and b.

.....

..... **[2]**

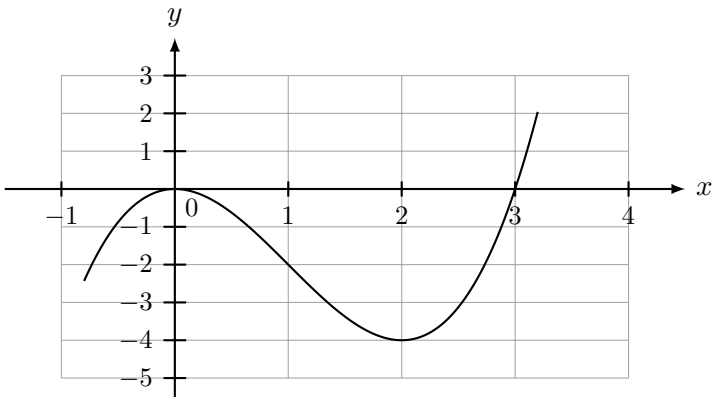
(b) Solve your equations to find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[2]**

(c) Find the coordinates of the other two points where the curve crosses the x-axis.

..... **[3]**

20 The diagram shows the graph of $y = f(x)$ where $f(x) = ax^3 + bx^2$.
The graph passes through the origin $(0, 0)$, the point $(3, 0)$ and has a local minimum point at $(2, -4)$.



(a) Show that $3a + b = 0$.

[1]

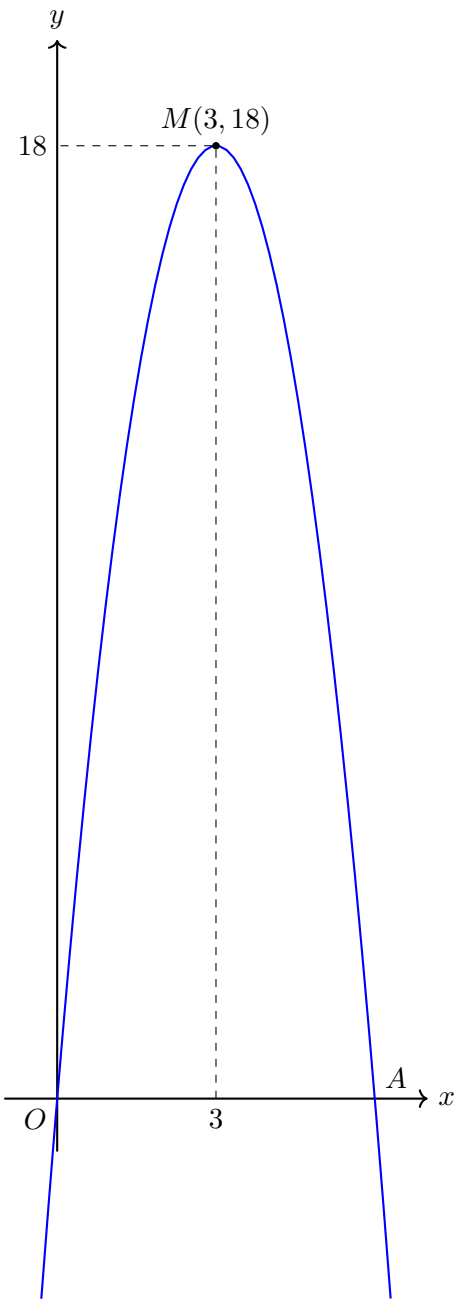
(b) Find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[3]**

(c) Write down the range of values of k for which the equation $f(x) = k$ has exactly one solution.

..... **[2]**

21 The diagram shows a sketch of the graph of $y = f(x)$, where $f(x) = ax^2 + bx + c$.
The curve passes through the origin O and intersects the x-axis at the point A.
The point $M(3, 18)$ is the maximum point of the curve.



(a) Write down the value of c.

$c =$ ..... **[1]**

(b) Write down the coordinates of A.

(..... , ...) **[1]**

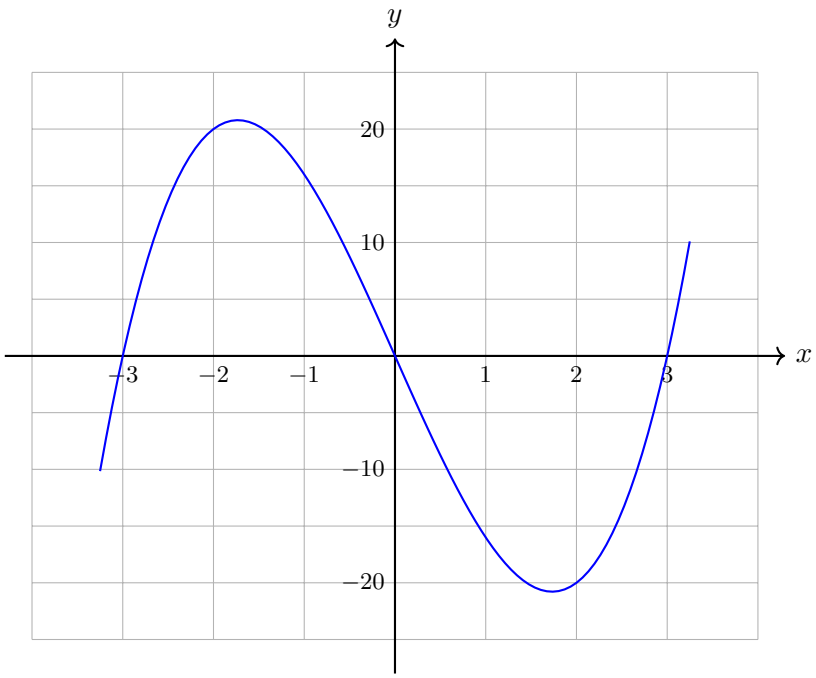
(c) Find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[3]**

(d) Find the range of values of k for which the equation $f(x) = k$ has two distinct positive real roots.

..... **[2]**

22 The diagram shows the graph of the cubic function $y = ax^3 + bx^2 + cx + d$.



(a) Write down the value of d.

$d =$ ..... **[1]**

(b) The graph intersects the x-axis at $x = -3$ and $x = 3$.
Show that $b = 0$ and $9a + c = 0$.

[3]

(c) The graph passes through the point $(1, -16)$.
Find the value of a and the value of c.

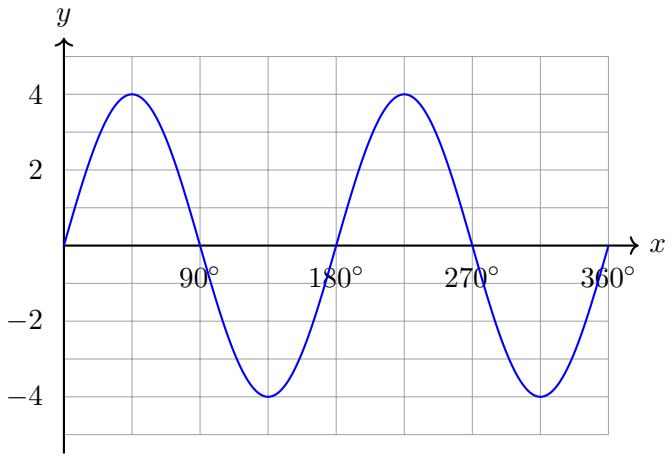
$a =$ ..... , $c =$ ..... **[3]**

(d) A different function $g(x) = px^2 + qx + r$ has a minimum point at $(0, -5)$ and passes through the point $(2, 7)$.

Find the values of p, q and r.

$p =$ ..... , $q =$ ..... , $r =$ ..... **[3]**

23 The diagram shows the graph of $y = a\sin(bx)$ for $0° \leq x \leq 360°$.



(a) Write down the value of a.

$a =$ ..... **[1]**

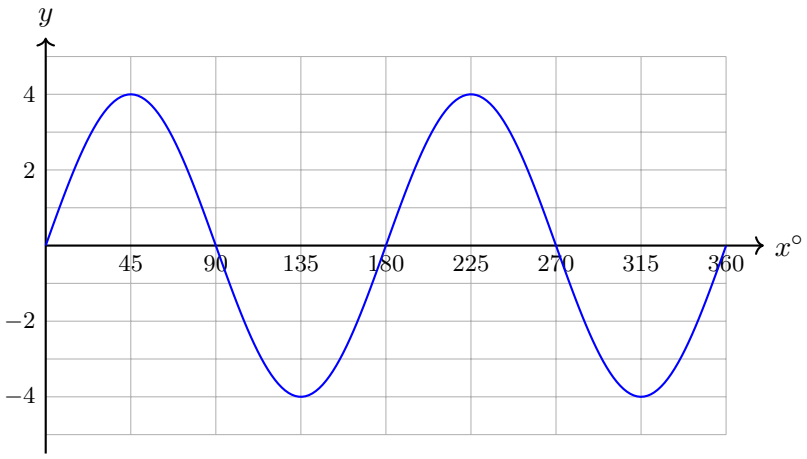
(b) Find the value of b.

$b =$ ..... **[1]**

(c) Calculate the value of y when $x = 30°$.
Give your answer correct to 3 significant figures.

$y =$ ..... **[2]**

24 The diagram shows the graph of $y = a\sin(bx)$ for $0° \leq x \leq 360°$.



(a) Find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[2]**

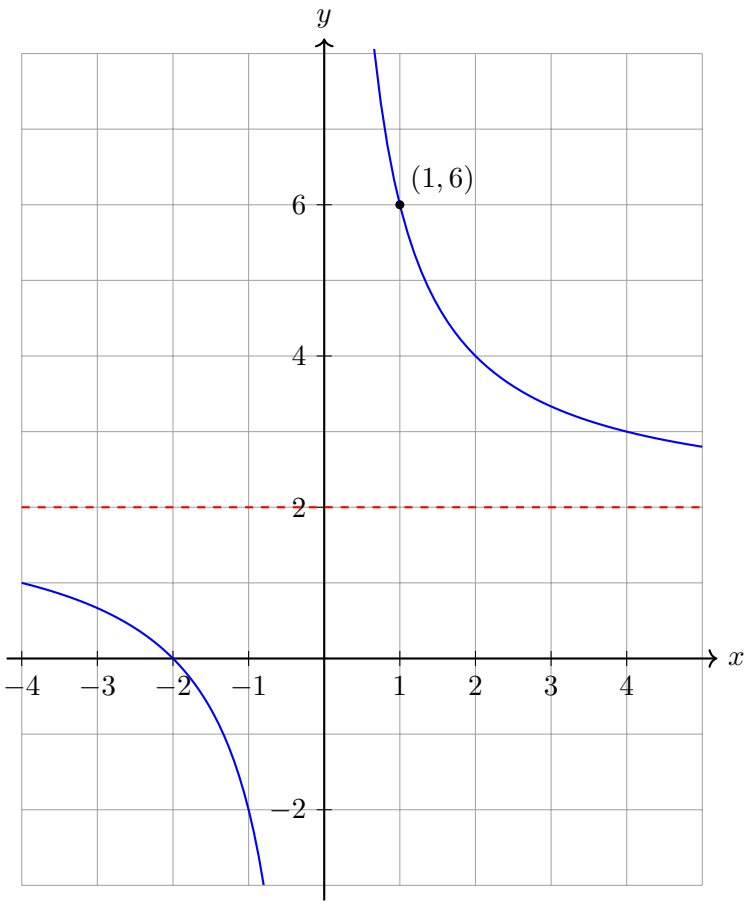
(b) The graph of $y = p^x$ passes through the point $(3, 125)$.
Find the value of p.

$p =$ ..... **[1]**

(c) The graph of $y = \dfrac{k}{x}$ passes through the point $(0.5, 8)$.
Find the value of k.

$k =$ ..... **[1]**

25 The diagram shows the graph of $y = \frac{a}{x} + b$.
The equation of the horizontal asymptote is $y = 2$.
The graph passes through the point $(1, 6)$.



(a) Write down the value of b.

$b =$ ..... **[1]**

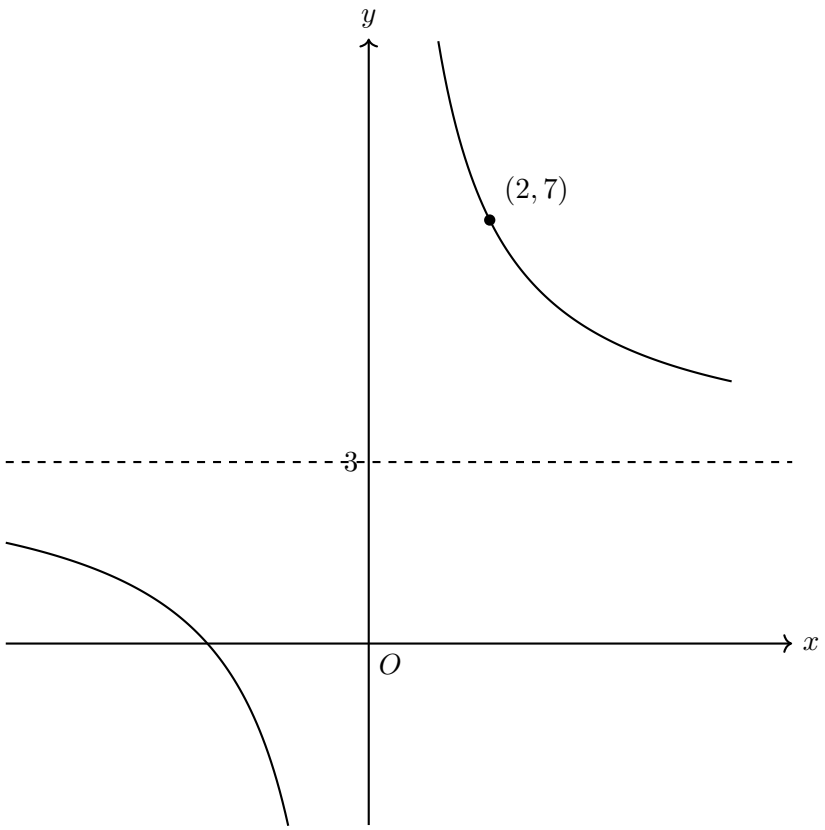
(b) Find the value of a.

$a =$ ..... **[2]**

(c) The graph of $y = k^x$, where $k > 0$, also passes through the point $(2, 4)$, which lies on the curve $y = \frac{a}{x} + b$.
Find the value of k.

$k =$ ..... **[2]**

26 The diagram shows a sketch of the graph of $f(x) = \frac{a}{x} + b$, where $xe0$.
The dashed line is a horizontal asymptote.
The graph passes through the point $(2, 7)$.



(a) Write down the value of b.

$b =$ ..... **[1]**

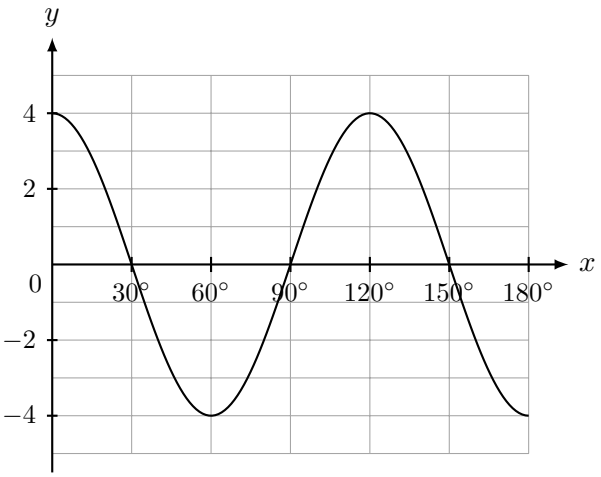
(b) Find the value of a.

$a =$ ..... **[2]**

(c) Find the coordinates of the point where the graph crosses the x-axis.

(..... , ...) **[2]**

27 The diagram shows a sketch of the graph of $f(x) = a\cos(bx)$ for $0° \leq x \leq 180°$.



(a) Write down the value of a.

$a =$ ..... **[1]**

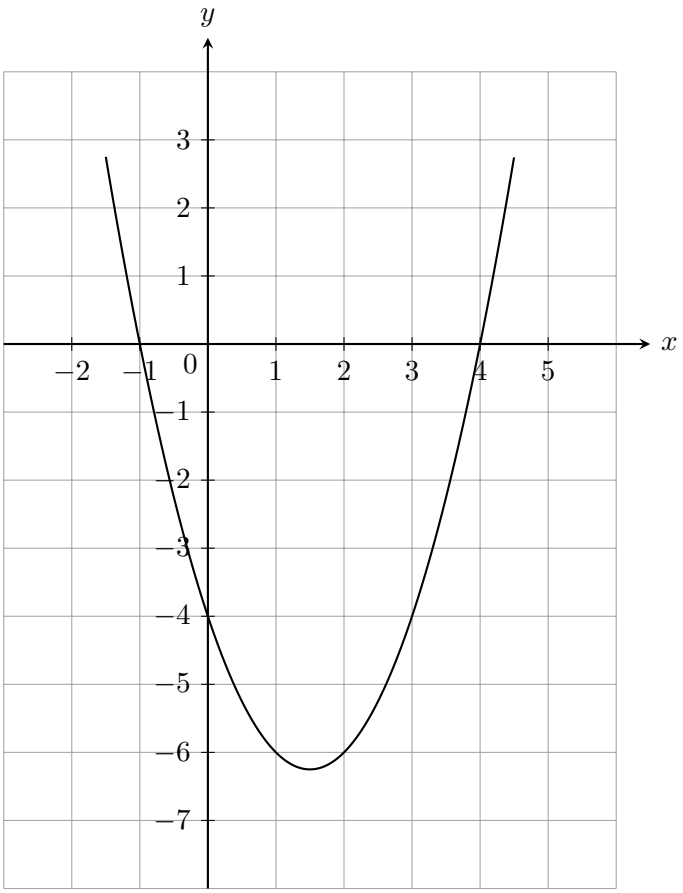
(b) Find the value of b.

$b =$ ..... **[2]**

(c) Determine the number of solutions to the equation $a\cos(bx) = 1.5$ in the interval $0° \leq x \leq 360°$.

..... **[2]**

28 The diagram shows the graph of a quadratic function $f(x) = ax^2 + bx + c$.
The graph crosses the y-axis at $(0, -4)$ and the x-axis at $(-1, 0)$ and $(4, 0)$.



(a) Write down the value of c.

$c =$ ..... **[1]**

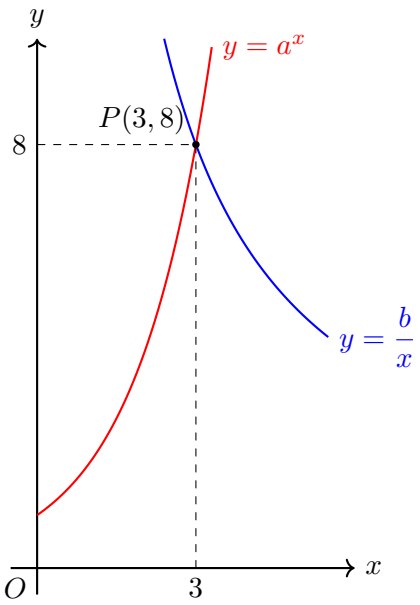
(b) Determine the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[3]**

(c) Calculate the exact coordinates of the minimum point of the graph.

(..... , ...) **[2]**

29 The diagram shows a sketch of the graphs of $y = a^x$ and $y = \dfrac{b}{x}$ for $x > 0$.
The graphs intersect at the point $P(3, 8)$.



(a) Find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[3]**

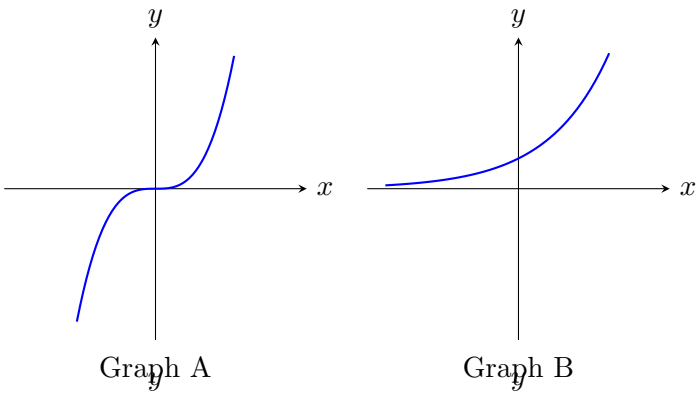
(b) The graph of $y = a^x$ intersects the y-axis at the point Q.
Write down the coordinates of Q.

(..... , ...) **[1]**

(c) Calculate the gradient of the straight line joining Q and P.

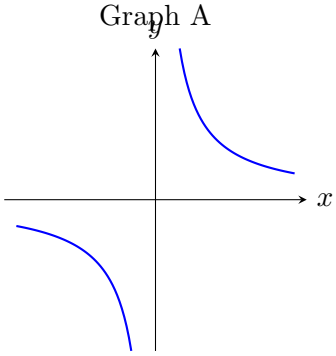
Gradient = ... **[2]**

30 Four graphs A, B, C and D are shown below.

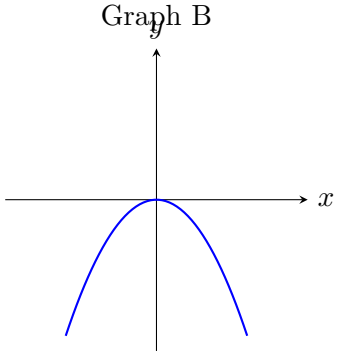


Graph A

Graph B



Graph C



Graph D

(a) Write down the letter of the graph which could represent the function

(i) $f(x) = \dfrac{a}{x}$ where $a > 0$

..... **[1]**

(ii) $f(x) = a^x$ where $a > 1$

..... **[1]**

(iii) $f(x) = -ax^2$ where $a > 0$

..... **[1]**

(b) The equation of Graph B is $y = p^x$.
The graph passes through the point $(3, 125)$.
Find the value of p.

$p =$ ..... **[2]**

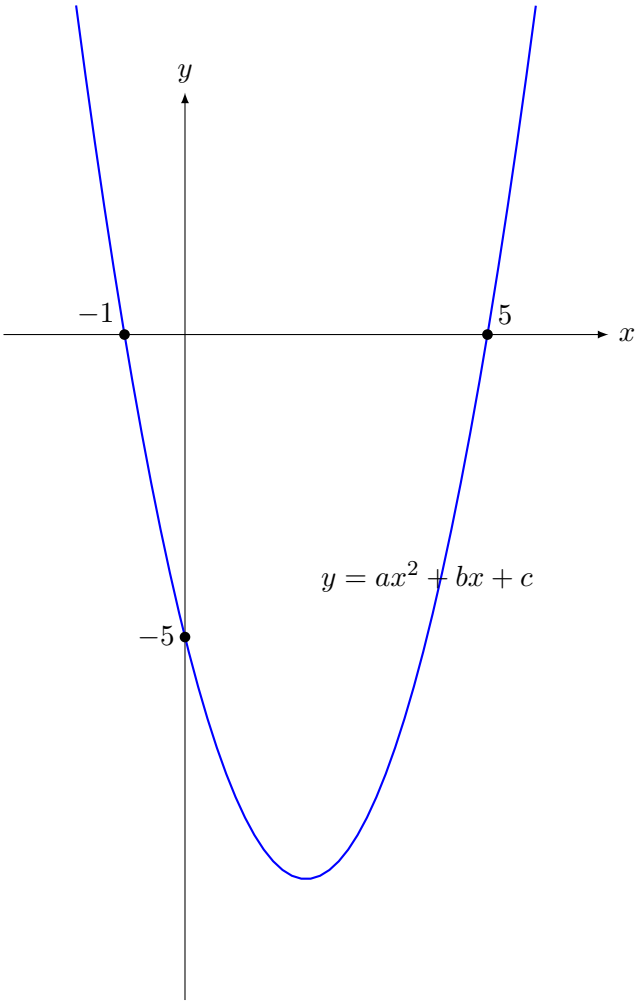
(c) The equation of Graph C is $y = \dfrac{q}{x}$.

The graph passes through the point $(0.5, 8)$.

Find the value of q.

$q =$ ..... **[2]**

31 The diagram shows the graph of $y = ax^2 + bx + c$.
The graph intersects the x-axis at $(-1, 0)$ and $(5, 0)$, and the y-axis at $(0, -5)$.



(a) Find the values of a, b and c.

$a =$ ..... , $b =$ ..... , $c =$ ..... **[3]**

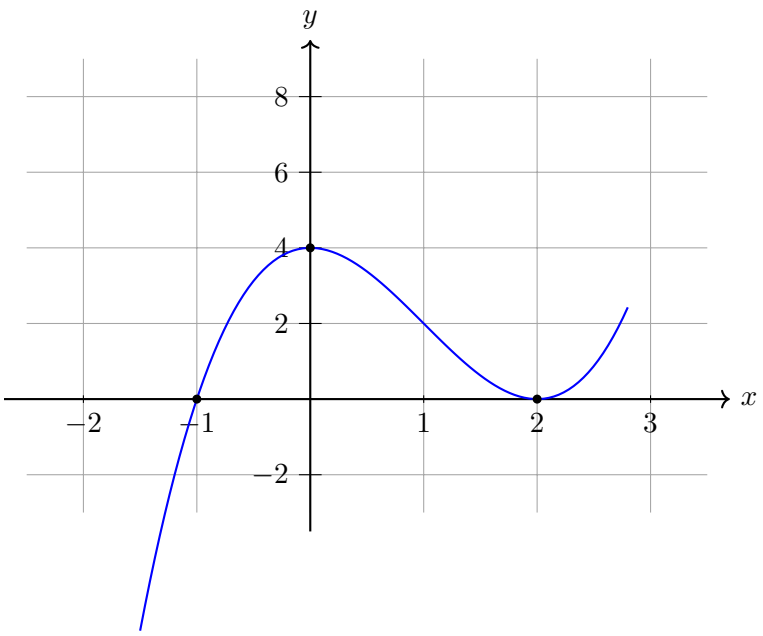
(b) Write down the coordinates of the minimum point of the graph.

(..... , ...) **[2]**

(c) The graph of $y = p \times q^x$ passes through the points $(0, 2)$ and $(2, 18)$, where $q > 0$.
Find the value of p and the value of q.

$p =$ ..... , $q =$ ..... **[3]**

32 The diagram shows a sketch of the curve $f(x) = x^3 + ax^2 + bx + c$.
The curve crosses the y-axis at $(0, 4)$.
It crosses the x-axis at $(-1, 0)$ and touches the x-axis at $(2, 0)$.



(a) Write down the value of c.

$c =$ ..... **[1]**

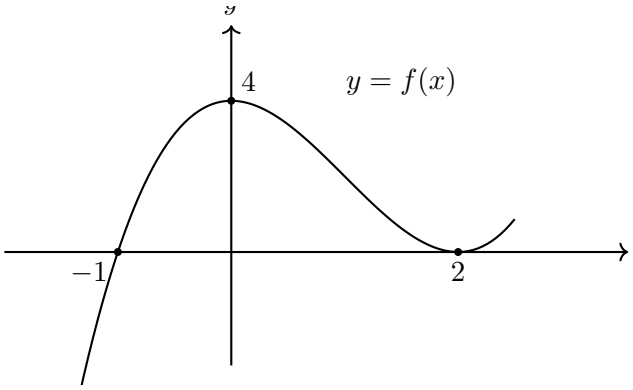
(b) By using the x-intercepts, find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[4]**

(c) Use the graph to state the number of solutions to the equation $f(x) = 3$.

..... **[1]**

33 The diagram shows a sketch of the graph of $f(x) = x^3 + bx^2 + cx + d$.
The curve intersects the y-axis at 4 and the x-axis at -1 and 2.
The point $(2, 0)$ is a local minimum.



(a) Write down the value of d.

$d =$ ..... **[1]**

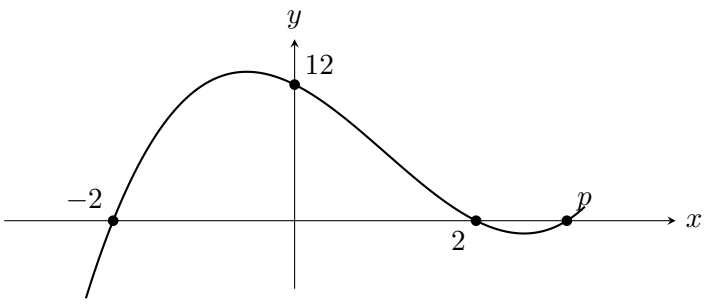
(b) Find the values of b and c.

$b =$ ..... , $c =$ ..... **[4]**

(c) Write down the number of solutions to the equation $f(x) = -3$.

Number of solutions = ... **[1]**

34 The diagram shows the graph of $y = x^3 + bx^2 + cx + d$.
The graph crosses the axes at $(-2, 0)$, $(2, 0)$, $(p, 0)$ and $(0, 12)$.



(a) Write down the value of d.

$d =$ ..... **[1]**

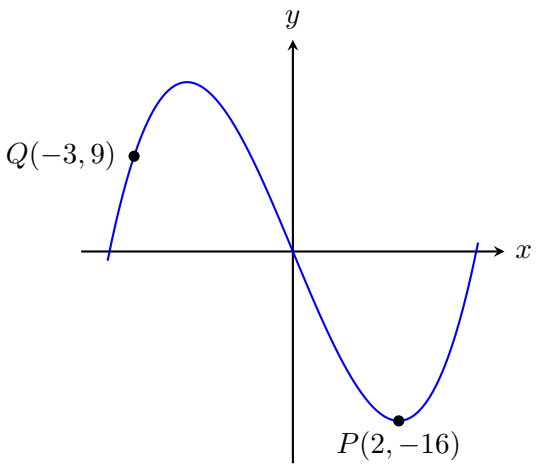
(b) Find the values of b and c.

$b =$ ..... , $c =$ ..... **[4]**

(c) Find the value of p.

$p =$ ..... **[2]**

35 The diagram shows a sketch of the graph of $f(x) = ax^3 + bx$.
The curve passes through the points $P(2, -16)$ and $Q(-3, 9)$.



(a) Find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[3]**

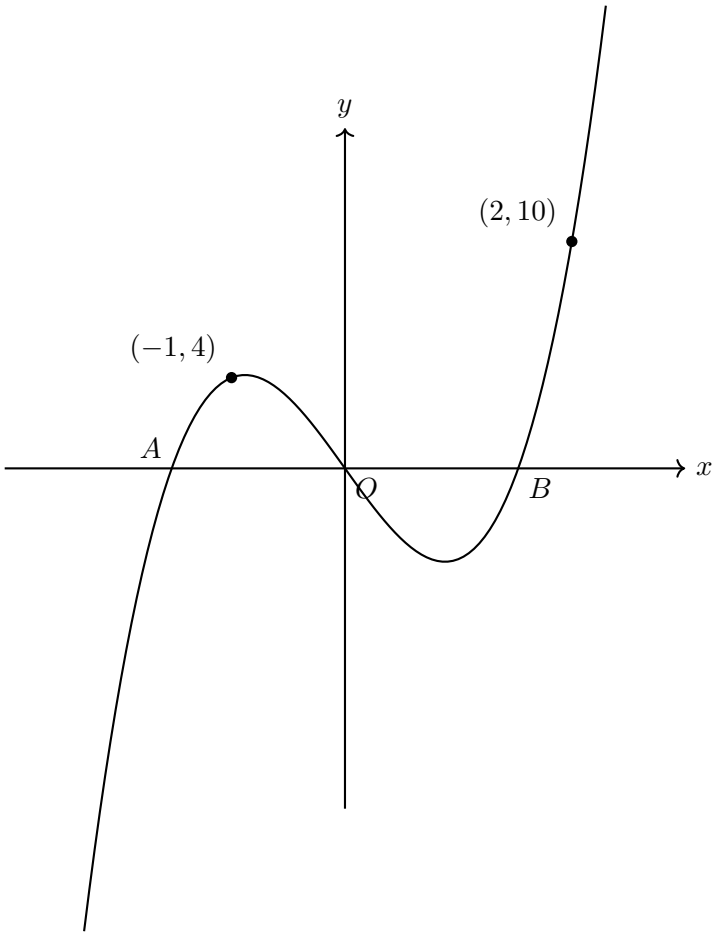
(b) Find the coordinates of the local maximum point of the graph.

(..... , ...) **[2]**

(c) The equation $f(x) = k$ has exactly one solution.
Write down the range of values for k.

..... **[2]**

36 The diagram shows a sketch of the curve $y = ax^3 + bx$.
The curve passes through the origin O, and the points $(2, 10)$ and $(-1, 4)$.
The curve crosses the x-axis at the points A, O, and B.



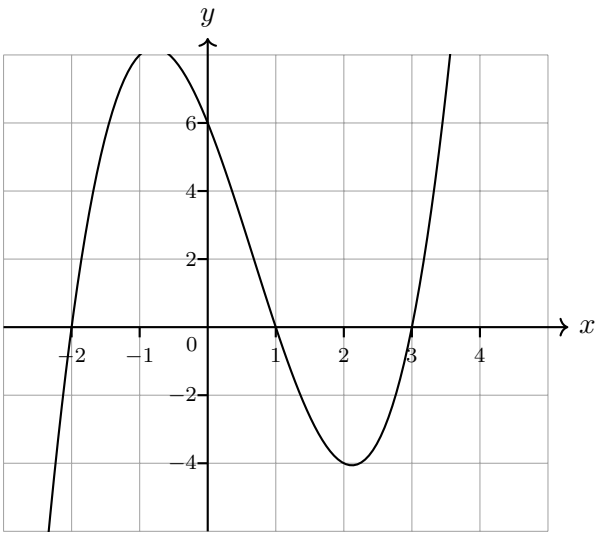
(a) Find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[4]**

(b) Calculate the exact length of the line segment AB.

..... **[3]**

37 The diagram shows the graph of the cubic function $g(x) = x^3 + bx^2 + cx + d$, where b, c, and d are integers.
The curve crosses the x-axis at $(-2, 0)$, $(1, 0)$, and $(3, 0)$.



(a) Find the values of b, c, and d.

$b =$ ..... , $c =$ ..... , $d =$ ..... **[3]**

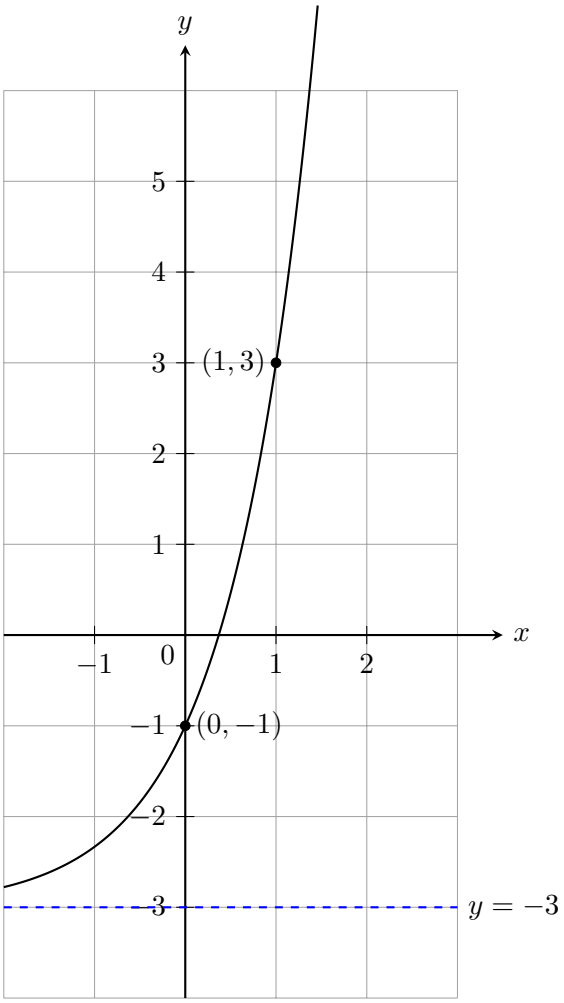
(b) Write down the coordinates of the y-intercept of the graph of $y = g(x)$.

(..... , ...) **[1]**

(c) Use your graphic display calculator to find the x-coordinates of the local maximum and local minimum points on the graph of $y = g(x)$.
Give your answers correct to 2 decimal places.

Local maximum $x =$ ..... , Local minimum $x =$ ..... **[3]**

38 The diagram shows the graph of the exponential function $f(x) = a \cdot b^x + c$.
The horizontal line $y = -3$ is an asymptote to the graph.
The graph passes through the points $(0, -1)$ and $(1, 3)$.



(a) State the value of c.

$c =$ ..... **[1]**

(b) Work out the value of a and the value of b.

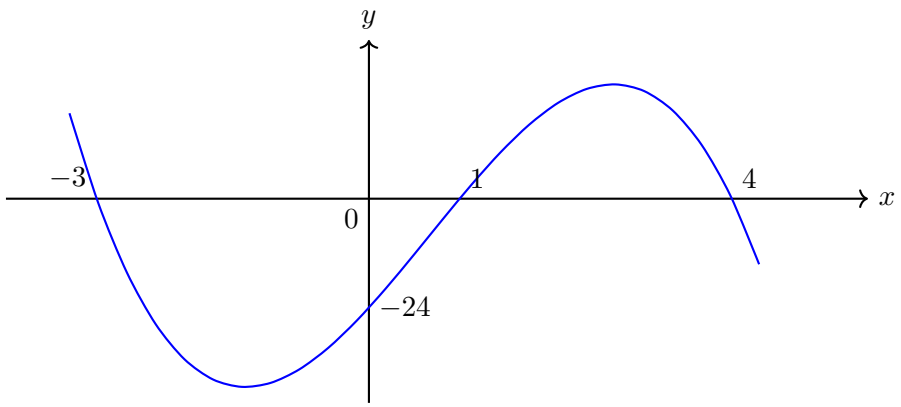
$a =$ ..... , $b =$ ..... **[4]**

(c) The graph of $g(x) = \dfrac{k}{x}$ intersects the graph of $f(x)$ at the point where $x = 1$.

Find the value of k.

$k =$ ..... **[2]**

39 The diagram shows the graph of a cubic function $f(x) = ax^3 + bx^2 + cx + d$.
The graph crosses the x-axis at $(-3, 0)$, $(1, 0)$ and $(4, 0)$.
It crosses the y-axis at $(0, -24)$.



(a) Find the values of a, b, c and d.

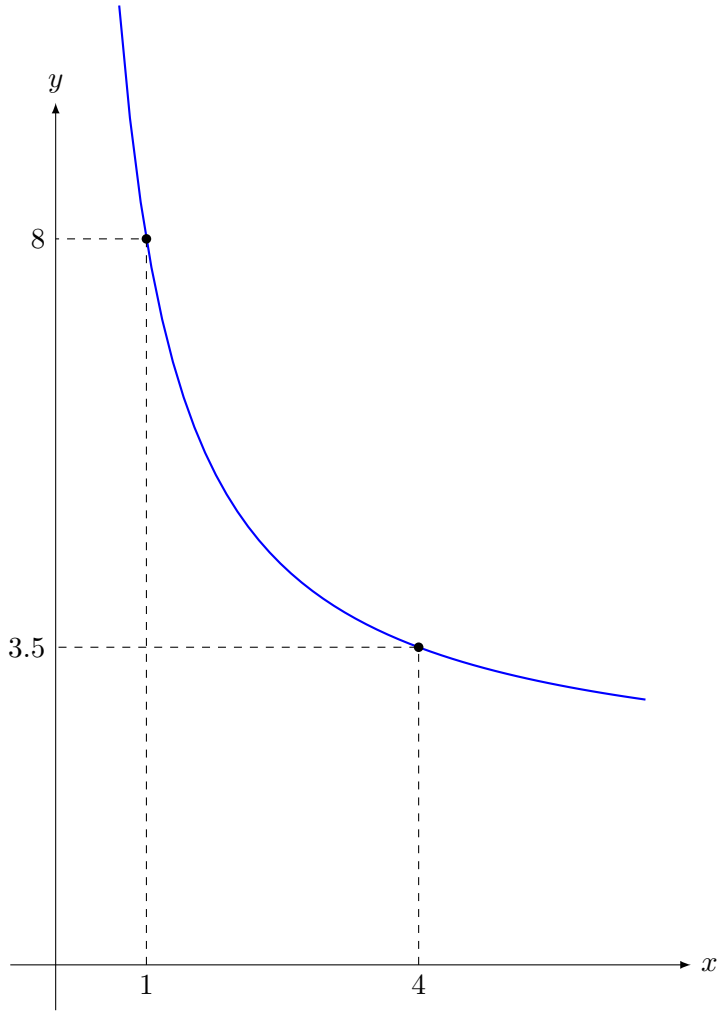
$a =$ ..... , $b =$ ..... , $c =$ ..... , $d =$ ..... **[6]**

(b) The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} 0 \\ 10 \end{pmatrix}$ to give the graph of $y = g(x)$.

Find the coordinates of the y-intercept of the graph of $y = g(x)$.

(..... , ...) **[2]**

40 The diagram shows part of the graph of $y = \dfrac{a}{x} + b$ for $x > 0$.
The graph passes through the points $(1, 8)$ and $(4, 3.5)$.



(a) Find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[3]**

(b) Write down the equation of the horizontal asymptote to the graph.

..... **[1]**

(c) The line $y = x - 3$ intersects the graph of $y = \dfrac{6}{x} + 2$ at the point P.

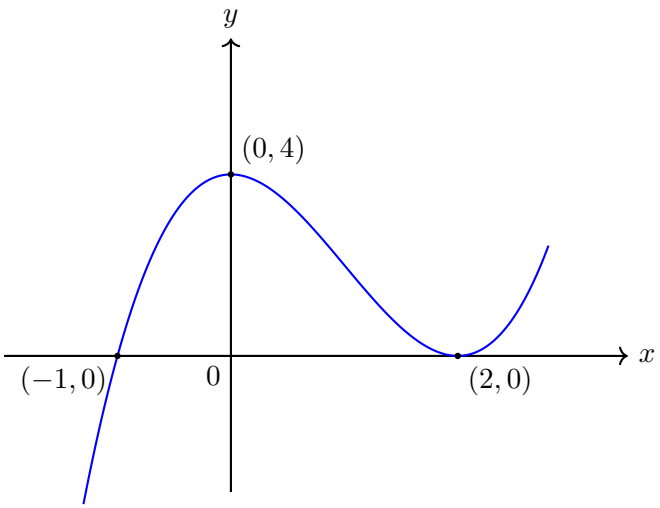
(i) Show that the x-coordinate of P satisfies the equation $x^2 - 5x - 6 = 0$.

[2]

(ii) Find the coordinates of P.

(..... , ...) **[2]**

41 The diagram shows a sketch of the graph of $f(x) = x^3 + bx^2 + cx + 4$.
The graph crosses the x-axis at $(-1, 0)$ and touches the x-axis at $(2, 0)$.
The graph crosses the y-axis at $(0, 4)$.



(a) (i) Use the point $(-1, 0)$ to show that $b - c = -3$.

[2]

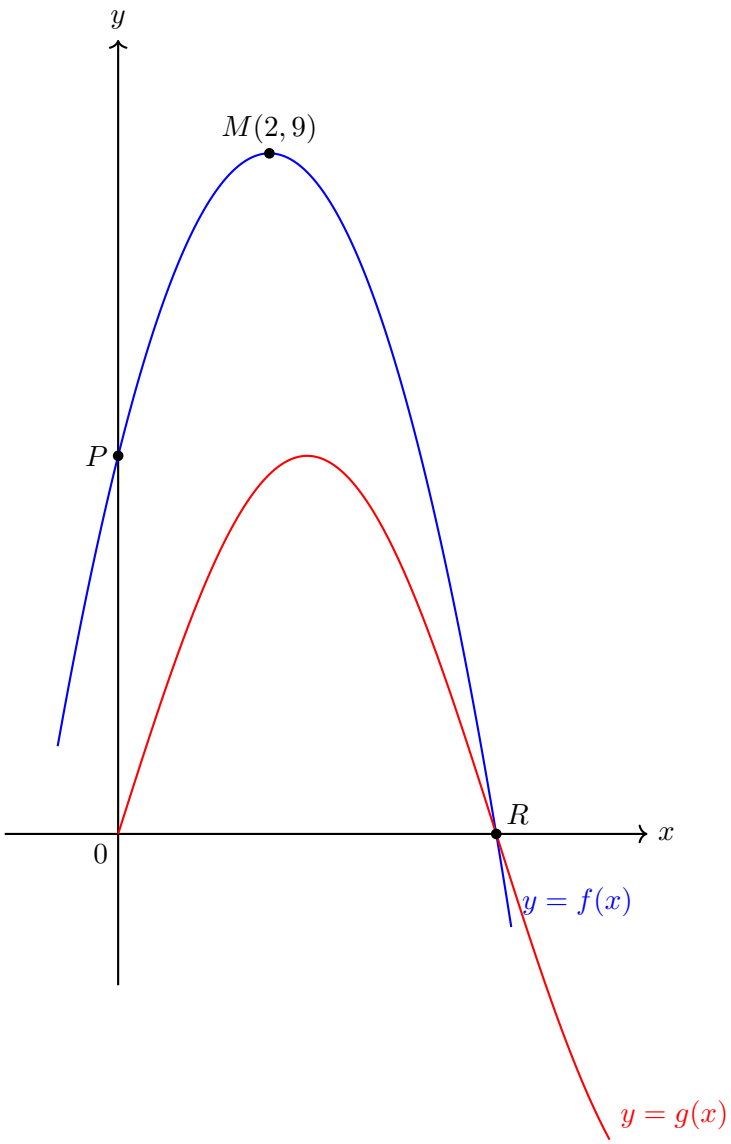
(ii) Find the values of b and c.

$b =$ ..... , $c =$ ..... **[4]**

(b) The graph of $g(x) = p \cdot q^x$ passes through the points $(0, 2)$ and $(3, 54)$.
Given that $q > 0$, work out the values of p and q.

$p =$ ..... , $q =$ ..... **[3]**

42 The diagram shows the graphs of $y = f(x)$ and $y = g(x)$ for $x \geq 0$.
It is given that $f(x) = px^2 + qx + r$.
The graph of $y = f(x)$ has a maximum point at $M(2, 9)$ and intersects the y-axis at $P(0, 5)$.



(a) Find the values of p, q, and r.

$p =$ ..... , $q =$ ..... , $r =$ ..... **[4]**

(b) The graph of $y = f(x)$ intersects the positive x-axis at the point R.
Find the coordinates of R.

(..... , ...) **[2]**

(c) It is given that $g(x) = a\sin(bx^\circ)$, where $a > 0$ and $b > 0$.
The graph of $y = g(x)$ has its first positive x-intercept at the point R.
The amplitude of $g(x)$ is equal to the y-coordinate of P.

Find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[3]**